

Chapter 9

Visualizing and Sonifying Mathematical Music Theory with Software Applications: Implications of Computer-Based Models for Practice and Education

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9.1. Introduction

This chapter discusses what software applications offer to the practice of mathematical music theory and, more broadly, to music education. Traditionally, mathematical music theory uses the formalisms of mathematics (such as equations and graphs) to represent and produce models of musical phenomena that serve explanatory, predictive, or generative purposes. By contrast, computer-based models use interactive elements to represent and model musical phenomena within a software environment.

These two approaches support quite distinct ways of working, thinking, and learning. The former entails a slow analytic process, constrained by step-wise workings and calculations, and

the latter allows for faster and more intuitive experimentation and hypothesis testing. While for most learners, learning to model musical phenomena in purely mathematical terms is cumbersome and challenging, well-designed software may enable such model-based learning to become more accessible and fluent. Thus, we argue that software can uniquely help visualize and sonify challenging concepts in mathematical music theory, helping students learn them more intuitively, and in ways that are more likely to support transfer to real musical situations.

It has been argued that computer-based modeling has constituted a major revolution in how people do science [36, 82] and how people learn challenging concepts in science and mathematics [25, 65, 66, 81]. A computer-based model, broadly defined, is a software application that allows the exploration of a mathematical or agent-based model computationally. This allows the computer to do the hard work on aspects of the modeling that are not directly relevant to learning or the conceptual domain in question (in our case: music theory). In other words, a computer is good at *scaffolding* our thinking and learning [71]. Another advantage is that computers are able to output real-time visualizations and sonifications of the model. For example, in science, one can run models on and visualize the dynamics of phenomena that are otherwise invisible, and thus difficult to learn and reason about, such as nanoparticles [43]. And so too in music, a computer-based model can output sonifications and visualizations of the musical structures we are modeling, which sometimes might be challenging to get a perceptual handle on.

How we represent musical information — whether it be with conventional notation, solfege, or mathematical formalisms — can have a significant effect on learning and practice [81]. Consider, for example, the historical shift from Roman numerals to the Hindu–Arabic numerals we use today. This change, far from being arbitrary, resulted in more people being able to understand and use mathematical ideas [6, 41, 81]. Similarly, we would like to propose that certain mathematical models and their instantiation in well-designed software are in a similar position to help more people learn

and understand music: a democratization of mathematical music theory.

In the next section, we consider some recent mathematical models of musical structure that can be usefully incorporated into interactive software applications and modeling activities such as those outlined above. We briefly summarize these theories and suggest some of the educational and creative affordances they provide. In the final section, Andrea Calilhanna — a practicing music teacher — will give a qualitative account of how introducing software based on mathematical representations positively influenced her teaching practice and her students' learning.

9.2. Mathematical Models of Musical Structure and Their Software Representations

Many recent mathematical models of musical structure are applicable to both the tonal and the metrical domains. This makes them particularly appealing in an educational context because any single such model has application across music's two most fundamental dimensions: pitch and time. Unlike conventional notation, which has different ways of representing pitch and time, mathematics provides a common vocabulary that invites learners to make connections between musical dimensions. And indeed there is a rich source of isomorphisms between pitch and time (e.g., Pressing [69], Monahan [60], though London [47] provides caution about extending the parallels too far). For example, both rhythms and scales are typically:

- periodic — for example, the diatonic scale repeats every octave; the clave rhythm repeats every two measures;
- subsets of a regular universe — for example, the diatonic scale is a 7-pitch subset of the 12-chromatic pitches; the clave rhythm is a 5-beat subset of 16 isochronous pulses;
- hierarchical — for example, the pentatonic scale is embedded in the diatonic which is embedded in the chromatic; a meter often

comprises numerous metrical levels each of which have differing groupings.

As subsequently detailed, notable recent mathematical music theories relevant to time, meter, pitch, and tonality include *maximal evenness* [14], *well-formedness* [11], *perfect balance* [51], *the discrete Fourier transform* [2], and *spatial isomorphisms* [19, 59]. This list is far from complete because the subsequent focus will be on mathematical models that are practical to visualize; for example, the unavoidably high dimensionality of voice-leading spaces (e.g., Tymoczko's voice-leading orbifolds) make them impractical unless restricted to two or, at most, three voices (David Bulger's useful interactive visualizations and sonifications of voice-leading spaces for two and three-part harmony, using Shepard tones, can be found at <http://science.mq.edu.au/~dbulger/>).

9.2.1. *Maximal evenness*

The property of *maximal evenness* [14] applies to periodic scales or rhythms where all pitches or beats fall at some regular division of the period (e.g., the octave or measure). For example, consider 7-tone scales in the familiar 12-tone equally tempered tuning; or 5-beat rhythms that fit into a grid of 16 *isochronous* (equally-spaced in time) pulses. When the number K of pitches in the scale divides into the number N of equally-spaced chromatic pitches, the maximally even scale is itself equally spaced, hence of limited interest (analogous results hold for rhythms).

But, when K does not divide N , the maximally even scale (or rhythm) always comprises no more than two step sizes (interonset intervals) and has a number of interesting properties: their complement is also maximally even; the two steps sizes are distributed as evenly as possible; every *generic* interval — e.g., second, third, fourth, etc. — comes in two *specific* sizes — e.g., m2 and M2, m3 and M3, P4 and Aug4; the distribution of sequences of specific sizes of seconds within larger generic intervals have the lowest possible entropy [54], which may confer cognitive advantages.

Furthermore, maximally even scales and rhythms are common in real-world music; for instance, the maximally even seven-pitch scale in 12-TET is the diatonic scale; its rhythmic analog is an archetypal West African timeline. The maximally even five-pitch scale in 12-TET is the anhemitonic pentatonic; perhaps the commonest of scales across the world. Numerous analogous examples of maximally even rhythms in world music are provided in [78].

The method to calculate maximally even scales (rhythms) is straightforward: Clough and Douthett [14] introduced a simple algorithm: for a K -pitch scale in an N -tone equal temperament, the octave is first divided into K equally-spaced pitches that are expressed in semitone units; then all these values are truncated to the nearest integer. But, performing calculations like this by hand, or with a spreadsheet, is tedious. Far better to allow users simply to choose an N and a K , and have the maximally even pattern automatically calculated, displayed and sonified (as in SequenceApp and Rhythm Necklace). Furthermore, multiple maximally even rhythms can be simultaneously sounded, each with differing rotations, allowing for maximally even polyrhythms.

Such maximally even structures, either expressed in pitch or time, are conducive to pedagogical treatments that teach mathematics with music and music with mathematics. Maximally even rhythms in cyclical graphs can be a grounding concept that can help children embody fundamental mathematical concepts and processes, including division and remainder, prime numbers, set inclusion, and modular arithmetic. Our research group is exploring such ideas in ongoing experiments, and we suggest that such cases present exciting opportunities for future work.

9.2.2. *Well-formedness*

There are many formal definitions of well-formedness [11, 83], all of which lead to the same end result. One way of understanding this principle is to consider it as a generalization of maximal evenness: a generalization that extends it from scales or rhythms with discrete values (i.e., subsets of a regular and finite universe of chromatic

pitches or isochronous pulses) to scales or rhythms that are subsets of a smooth continuum. This generalization allows scales whose pitches do not fall at some equal division of the octave (e.g., the historically important quarter-comma meantone and Pythagorean tunings [9]; or rhythms, such as swung jazz rhythms, that use consistent microtimings to impart “groove” [28]).

In a smooth continuum, the maximally even scale or rhythm will always be equal-stepped (isochronous) because, unlike in the discrete context, the pitches (beats) are no longer constrained to fall at a specific N th subdivision of the octave or metrical period. But, a well-formed scale (rhythm) works with an alternative constraint: it is the maximally even scale or rhythm comprising a given number of *large* and *small* steps. For example, the well-formed scale with five large steps and two small steps is the diatonic scale. But, unlike the maximally even version, the precise sizes of these scale steps are now flexible.

This means that the diatonic scale is well-formed regardless of its precise tuning — it could be in 12-TET, it could be in 17-TET, or 31-TET, or 19-TET, or it could be in a tuning with no equal steps at all, such as the previously mentioned meantone and Pythagorean tunings. All of these well-formed diatonic scales have the same pattern of large (L) and small (s) steps ($LLsLLLs$), but the ratio of the sizes of L and s differs. For example, in the 12-TET diatonic, $L/s = 2$; that is, the large steps (major seconds) are twice the size of the small steps (minor seconds); in the 19-TET diatonic, $L/s = 3/2$; in the quarter-comma meantone diatonic scale, $L/s = \log(25/16)/\log(4096/3125) \approx 1.649$.

A similar generality across scale-step ratios (or interonset interval ratios) holds for scales and rhythms with any pair of numbers of large and small steps; for example, 4 large steps and 3 small, or 3 large steps and 5 small. Each possible coprime pair of these numbers produces a unique family of new scales and rhythms. And all of them have properties analogous to those of maximally even scales or rhythms: their interval size distribution minimizes entropy, every generic interval comes in no more than two specific sizes, and so forth.

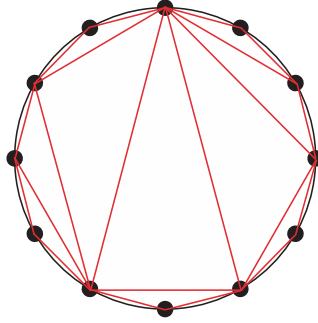


Fig. 9.1. The well-formed hierarchy of perfect fifth, tetractys, pentatonic, diatonic, chromatic (detailed in the main text).

But, well-formed scales (rhythms) also possess a number of additional useful properties, as detailed in Milne *et al.* [53], and Milne and Dean [54]. Notable amongst these being their hierarchical nature; for example, the principal scales in Western music are all well-formed and all hierarchically “nested” sets: the well-formed perfect fifth (D, A) is a subset of the well-formed perfect fifth (D, A) is a subset of the well-formed tetractys (D, G, A), which is a subset of the well-formed pentatonic (C, D, E, G, A), which is a subset of the well-formed diatonic (C, D, E, F, G, A, B), which is a subset of 12-pitch class chromatic set (see Fig. 9.1, for a visualization of this hierarchy). Hierarchical embeddings such as these are particularly useful for rhythms and meters, which are commonly hierarchical in structure.

The broad generality of this model of scale and rhythm construction has been exploited in the Dynamic Tonality software developed by Milne and colleagues (<http://www.dynamictonality.com>). In terms of pitch, the MIDI sequencer Hex [68] allows music to be composed in arbitrary well-formed scales that can be smoothly changed in tuning, and a suite of synthesizers can sonify these musical scales and produce matched timbres to maximize perceived consonance [55, 57, 58, 74]. In terms of rhythm, the Dynamic Tonality loop generator XronoMorph facilitates the creation and manipulation, in real time, of hierarchical well-formed rhythms

and melodic hockets [55, 57]. This produces rhythms that, while remarkably complex, are still groovy; furthermore, different rhythms can be smoothly morphed between — a novel and striking musical effect. Video demonstrations of XronoMorph can be viewed at <https://www.youtube.com/c/xronomorph>.

The above applications, and enthusiastic responses to them from musicians and music producers, have demonstrated their potential for facilitating creative engagement with, at times, unorthodox musical structures that break conventional constraints on tuning or timing. In terms of music education, they can provide effective examples of how extracting features from traditional systems — such as the pentatonic–diatonic–chromatic hierarchy — and recasting them into different, but analogous, forms can generate interesting and novel variations.

9.2.3. *Perfect balance*

Perfect balance [51, 52] is a recently developed organizational principle for periodic scales and rhythms. It is conceptually simple: if pitch classes or beat classes are represented by point masses on a circle, the resulting scale or rhythm is perfectly balanced if their centre of mass (mean position) is at the centre of the circle. More generally, the *balance* of a scale or rhythm is the proximity of its centre of mass to the centre of the circle, normalized so that a balance of 1 implies perfect balance, and 0 implies perfect imbalance (which occurs only when all pitches are the same, or all beats occur simultaneously).

All isochronous rhythms are perfectly balanced and the sum of such isochronous rhythms is also perfectly balanced. This, therefore, includes most conventional metrical structures (like 3/4, 4/4, 5/4, 6/4, . . . , 3/8, 5/8, 6/8, 7/8, . . .) as well as familiar polyrhythms — common in African musical traditions — where a period is divided into both 2 and 3, or into both 3 and 4, or into both 3 and 5, and so on. Furthermore, it also includes polyrhythms where the different cycles are out of phase so they never coincide (a musical possibility perhaps first identified by Hofstadter [33]); perfectly balanced *interweaving polyrhythms* [4], such as these, are also found in African musical traditions [51] (Fig. 9.2).

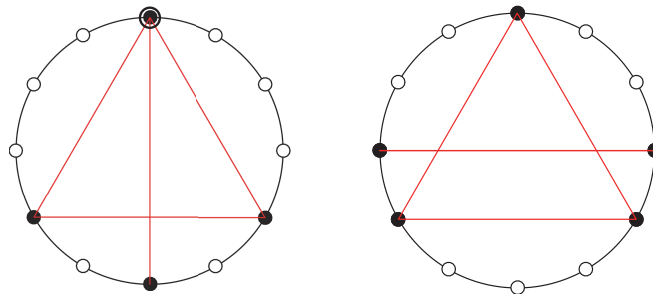


Fig. 9.2. Two against three polyrhythms: the first is the classical hemiola, the second is an interesting variant where the two rhythms are out of phase and never coincide. Both are perfectly balanced.

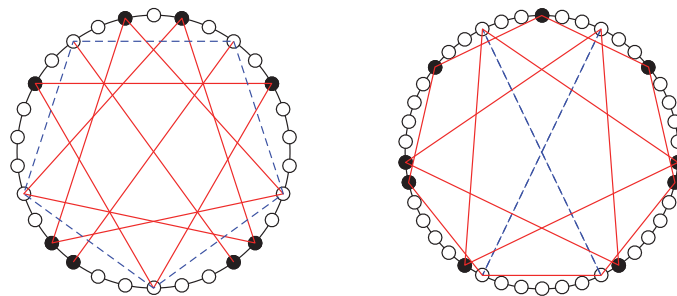


Fig. 9.3. Two minimal perfectly balanced primitive rhythms. The regular polygons with dashed lines are negatively-weighted; the regular polygons with solid lines are positively-weighted. When vertices of a positive and negative polygon coincide, they cancel each other out. Minimal perfectly balanced primitive patterns, such as these, cannot be constructed solely from positively-weighted regular polygons [51]. Perfectly balanced rhythms such as these can be explored — visualized and sonified — in XronoMorph (see also <https://youtu.be/uKcOOBGxzJc>).

Intriguingly, there are further generalizations of polyrhythms opened up by the mathematical definition of perfect balance (Fig. 9.3). Instead of simply *adding* isochronous beats, it is also possible to *subtract* isochronous beats. This results in meaningful rhythms so long as the result has no negatively-weighted beats. In this way, “ghostly” polyrhythms can be created where the impact of the subtracted rhythms is the absence rather than the presence of a sounded beat. Figure 9.3 shows the construction of two of these *minimal primitive perfectly balanced* rhythms.

From the perspective of teaching music, these visualizations with regular polygons make the construction of polyrhythms clear and easy to understand. This is important because representing polyrhythmic structures either with conventional notation or on a MIDI pianoroll, obfuscates their underlying simplicity. The generalizations of perfect balance, also allow for unusual, but musically interesting, polyrhythms to be explored. From the perspective of teaching mathematics, these patterns can help learning the concept of coprimeness: any number of regular polygons with coprime vertices can only ever all meet (visually and sonically) at one point in the period; when the numbers of vertices are not coprime, they can intersect at multiple points. One might imagine a group of students — guided by the animated visualization in the software — tapping out polyrhythms to fully embody this understanding.

9.2.4. *DFT coefficients*

A discrete scale can be conveniently represented by an *indicator vector*; for instance, the diatonic scale can be represented by the 12-entry indicator vector $(1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 1)$, where the ones represent the presence of a pitch in the scale and a zero represents the absence of a pitch in the scale. The discrete Fourier transform of this vector is another vector of 12 elements, each of which is a complex number. The magnitudes and phases of the first seven coefficients (more, generally $\lceil N/2 \rceil$, where N is the number of chromatic pitches or isochronous pulses) can be used to identify and manipulate interesting properties of the scale. This use of the discrete Fourier transform was introduced by Lewin [46], and exploration of the meanings of these coefficients has been a hot topic in recent music theory [1, 2, 10, 27, 70].

Indeed, the magnitude of the first coefficient is proportional to balance (as defined above). In the 12-TET context, the magnitude of the fifth coefficient has been characterized as the “diatonicity” of the scale; the magnitude of the fourth coefficient as its “octatonicity”; the magnitude of its sixth coefficient as its “whole-tone quality”. Clearly, with different N , such as in a microtonal scale system or

in many rhythms, the musical meanings of the coefficients (with the exception of the first) differs.

One utility of the discrete Fourier transform is that it transforms the common musical — but mathematically difficult — operation of convolution into the simpler operation of multiplication (indeed it is the only linear transformation that can do this [2, Theorem 1.11]). Convolution in the pitch domain corresponds to the musical operation Cohn [17] termed *transpositional combination*: it is the process by which a pitch class set is transposed and then added. Such scales are *rotationally symmetric* (they are also known as transpositionally invariant or TINV pitch class sets) and have found frequent use and been met with much theoretical interest; for example, all of Messiaen’s modes of limited transposition [49] are rotationally symmetric [16]. Temporal analogs for rhythm also occur [17]. Amiot [2, p. 7] suggests that the discrete Fourier transform is “unreasonably efficient” in music theory.

From the creative perspective, *Fourier scratching* is a technique developed by Amiot, Noll, Andreatta, Agon, and Carlé [3, 12, 53] where musical outputs can be controlled directly by manipulating the coefficients of a scale’s discrete Fourier transform. It seems likely that useful software interfaces and realtime visualizations will also be developed for educational and analytical purposes. The musical illustrations may also provide motivation for more advanced students to better understand the Fourier transform.

9.2.5. *Interactive spatial isomorphisms of musical structures*

Spatial isomorphisms help to make visualizations of pitch and tonality, time and meter, and their underlying mathematical constructions, easy to understand and to manipulate. In a musical context, the term “isomorphism” is usually applied to two-dimensional layouts of pitches in musical interfaces; but, as shown subsequently, it also applies to higher order musical structures such as tonality and meter.

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The definition of isomorphism relevant here (this is not a strict mathematical definition) is a one-to-one and invertible (structure-preserving) map between two domains. Here, the domains are some musical feature (such as pitch, metrical level, chord) and space (one, two, or three-dimensional). The focus, here, will be on two-dimensional space, because this affords more interaction possibilities than a sole dimension, and because it is more practical to implement than three dimensions. And because isomorphic spatial representations of musical structures are structure-preserving and consistent, it is often implied that they are intuitive and easier to understand than non-isomorphic representations.

The importance of considering spatial information and the extent of its isomorphism, or congruity, with musical information is emphasized by a growing body of cognitive and neuroscientific research showing that the way we process space is closely connected with how we process the musical dimensions of both pitch [22, 26] and time [37]. And, more generally, similar patterns are found in other domains such as mathematics [40, 44, 45, 50] and scientific reasoning [24]. Some go as far as to say that space is at the very heart of human thought [42].

The interactivity of representations also matters. If we accept that space can be meaningful, we interface with and manipulate this meaning through action. For example, Goldstone, Landy and Son [29] studied how experts in algebra solved equations such as $2x + 5 = 13$. Rather than abstractly computing the correct answer, like a computer would, they found evidence to support the idea that experts mentally visualize the mathematical symbols in their mind's-eye and interact and manipulate the symbols with imagined gestures, such as dragging the '5' to the other side of the equation. Based on this, they designed a software application called Graspable Math [30, 62] that supports the development of this dynamically embodied way of thinking about algebra — <https://graspablemath.com/>. The software supported this by having the symbolic representations of mathematics dynamically respond to gestures such as dragging.

We will now provide an overview of representations in music that are spatially isomorphic with musical structures and offer affordances for meaningful interaction and gesture.

9.2.5.1. *Pitch layouts*

Physical isomorphic pitch layouts (also known as *generalized keyboards*) date back to the nineteenth century, with numerous examples detailed by Helmholtz and Ellis [32] (see also Keislar [39]). These were designed to make conventional music easier to play than on the conventional piano keyboard by ensuring that the same musical interval always has the same “shape” (i.e., angle and distance) regardless of transposition. For example, the major third between C and E has the same shape as the major third between D and F♯ (something that is not true of the piano keyboard). They were also designed to facilitate the use of musical tunings that more closely approximate just intonation than does 12-TET (e.g., 53-TET and 31-TET). Numerous recent hardware musical controllers have also utilized isomorphic layouts: Array Mbira [5], Thummer [63], AXiS-49 [23], LinnStrument [72], Lightpad Block [73], and Terpstra [77].

An advantage of software interfaces combined with multitouch screens is that they easily allow for a variety of such layouts to be explored; for example, the iOS app Musix Pro [67] provides a range of isomorphic layouts for 12-TET, while Hex [68] uses such layouts for MIDI sequencing.

It has been argued that isomorphic layouts have important benefits for elucidating the structure of tonality. For example, they make explicit the invariance of intervallic structure when transposing to a different key. They also clearly expose the structure of well-formed scales (defined below) such as the pentatonic and diatonic, because these are always spatially represented as notes arranged on successive lines at different heights — see the “3 + 2” pattern of rows made by the pentatonic scale and the “3 + 4” pattern made by the diatonic scale in Fig. 9.4.

Furthermore, every isomorphic layout has two *axes* that indicate two important musical dimensions: *pitch height* and *perfect fifth*

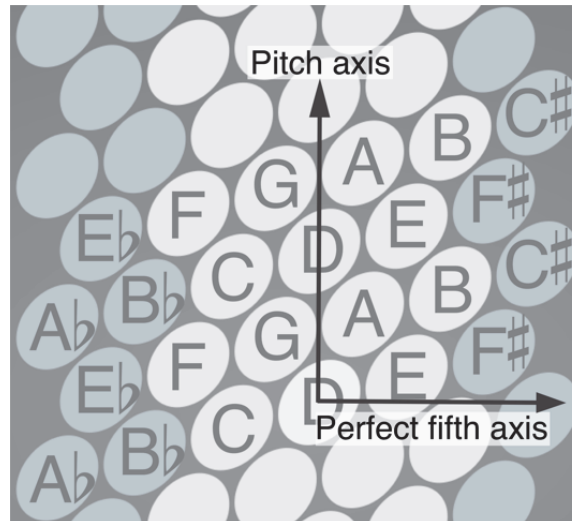


Fig. 9.4. A pitch layout where the pitch axis is vertical and the perfect fifth interval class axis is horizontal, making both of these important dimensions of pitch distance easy to see or to sense by proprioception. The well-formed pentatonic scale follows a 3 + 2 pattern of rows, the well-formed diatonic scale follows a 3 + 4 pattern of rows. In microtonal tunings, related patterns occur for all possible well-formed scales.

interval class “width” (IC5 width, for short). The *pitch axis* is any line angled so that, along this line, the position of every pitch-playing button-centre is proportional to its pitch height; for instance, the pitch layout in Fig. 9.4 has a vertical pitch axis, hence the vertical height of every pitch-playing button corresponds to its pitch height. The *IC5 axis* (termed the *generator span axis* in [68]) is any line angled so that, along this line, the position of every pitch-playing button-centre is proportional to its position in the chain of perfect fifth interval classes (IC5s); for example, the pitch layout in Fig. 9.4 has a perfectly horizontal IC5 axis, hence the left–right position of any button is proportional to its position in the chain of fifths. These two aspects of tonal relations — pitch height and the chain-of-fifths (or circle-of-fifths, depending on the precise tuning used) — are salient features of Western music and its theory [13, 31].

This makes the precise pitch layout shown in Fig. 9.4 interesting because its pitch axis and its IC5 axis are orthogonal and perfectly vertical/horizontal (this orthogonality is achieved by geometrically shearing a standard [79] layout). In recent experiments, we have established that having a perfectly vertical pitch axis strongly facilitates the learning and playing of melodies (the impact of the angle of the IC5 axis has not yet been tested) [48].

9.2.5.2. Chord/key tonnetz

When considering relationships between chords or keys (as represented by their tonic triad), a different isomorphic representation — based on pitch classes rather than pitches — has proved useful: the *chord Tonnetz*. Although commonly represented in a regular hexagonal form (see Fig. 9.5(a)), experiments obtaining ratings of the fits of different pitches and chords have shown that a linear

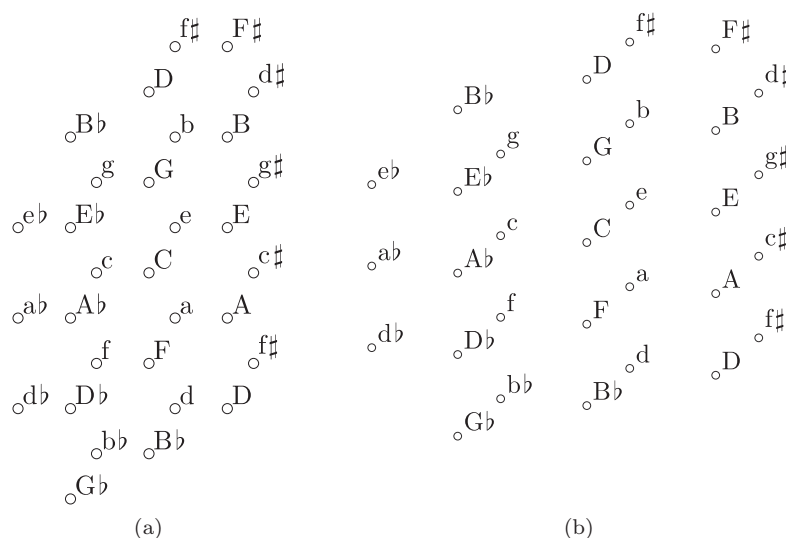


Fig. 9.5. The chord Tonnetz on the left is in the classical regular hexagonal form (major triads are in upper case, minor triads in lower case). The chord Tonnetz on the right has been sheared and scaled to optimize the correlation between spatial distances and ratings given by 42 participants of the similarity and fit of all 26 possible pairs of root position major and minor triads (the resulting correlation is a substantial $r = 0.92$ [56]).

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transformation (involving a shear and a one-dimensional scaling) of the regular form provides a substantially better fit of spatial distances to perceptual distances [56] — as shown in Fig. 9.5(b).

It seems clear that the chord Tonnetz has potential for allowing music students to quickly visualize not just the relationships between chords, but also the degree of fit between pairs of such chords — the manipulation of fit and tension being, perhaps, one of the principal devices used in tonal composition for the expression of emotions.

The power this brings, like with pitch layouts, is afforded by the cognitively congruent use of spatial information. Just as displaying algebra in spatially congruent ways facilitates mathematical thinking [45], it can be said here that displaying tonal information in spatially congruent ways supports musical thinking. The effectiveness of such a representation may be further extended by rendering it in software. A computer-based model of this could sonify the tonal relationships that are expressed spatially. This would enable a qualitatively different modeling process involving intuitive back-and-forth interaction between ideas suggested by the spatial information and their auditory consequences. One such application is Harmony Space, which is a tool designed to facilitate rapid learning about the practical use and theory of tonal harmony [34, 35]. Such software has applications for both teaching and creative composition.

9.2.5.3. *Circles, polygons and periodicity*

A natural isomorphic representation of the periodicity found in musical scales and rhythms is a circular form (as in Figs. 9.6–9.8).^a The software applications XronoMorph (see Fig. 9.6), Rhythm Necklace, SequenceApp, and Euclidean Pattern Generator all represent

^aHaving said that, in many situations (particularly melodic), pitches an octave apart are not perceived as identical. A common model of pitch perception is as a helix, where pitch class (chroma) is the circular dimension and pitch height is the orthogonal linear dimension [75]. Using Shepard tones, which comprise partials that are only in octave relationships, makes it possible to seamlessly sonify just this circular dimension. For rhythms, the circularity is easier to make concrete because special sounds are not required.

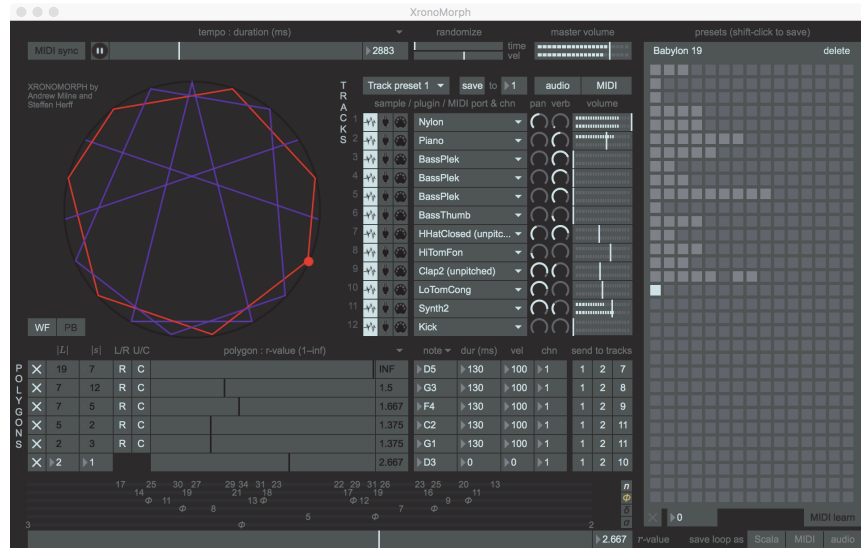


Fig. 9.6. The user interface of XronoMorph showing, on the left, a circular representation of a multilayered (hierarchical well-formed) rhythm. Each rhythmic level is distinguished through the use of inscribed polygons. One advantage of keeping all rhythmic levels on the same periodic circle is that the temporal and geometrical relationships between them can be easily seen.

rhythmic loops in a circular format, with periodic time mapped to angle. XronoMorph and some of the others also use polygons created by connecting together the rhythmic points on the circle (i.e., the polygons are inscribed in a circle). These cyclical graphs can be a useful way of visually differentiating between multiple voices — such as may be found in a polyrhythm — on the same circle.

Rendering a cyclical graph in software can also enable meaningful embodied interactions such as those supported for algebra in Graspable Math. For example, consider the concept of transpositional equivalence in set theory: to obtain a transposition by 2 of the pitch-class set $\{0, 4, 7\}$ (a major triad), one may follow the canonical algorithm of adding 2 to each member to reach $\{2, 6, 9\}$. However, a more cognitively informed approach would be to conceive of the transpositional relationship as ‘rotational equivalence’; that is, as a perception–action routine. The polygons represented

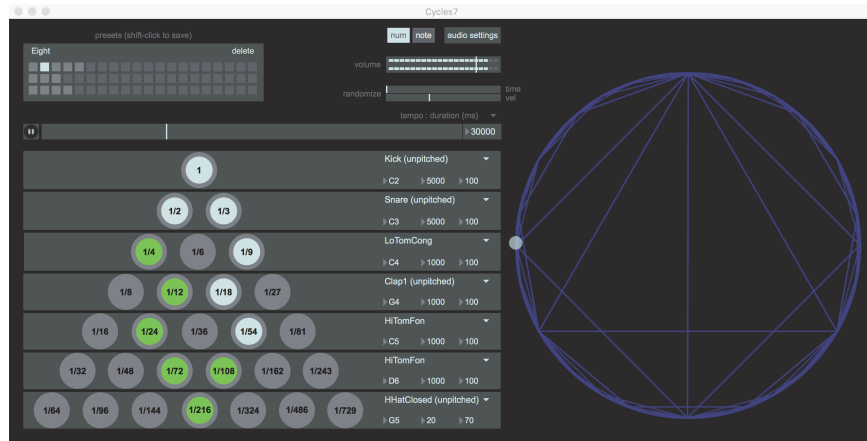
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Fig. 9.7. A prototype of the SkiHill software application — discussed, later, by Andrea Calilhanna. A user can select any node to make its corresponding pulsation sound — a selected node is highlighted and it flashes green while actually sounding. Each row (metrical level) can be assigned to an instrumental sound, so that the whole metrical structure can be sonified by a virtual percussion ensemble. Like XronoMorph, the rhythm is also illustrated with polygons inscribed in a circle.

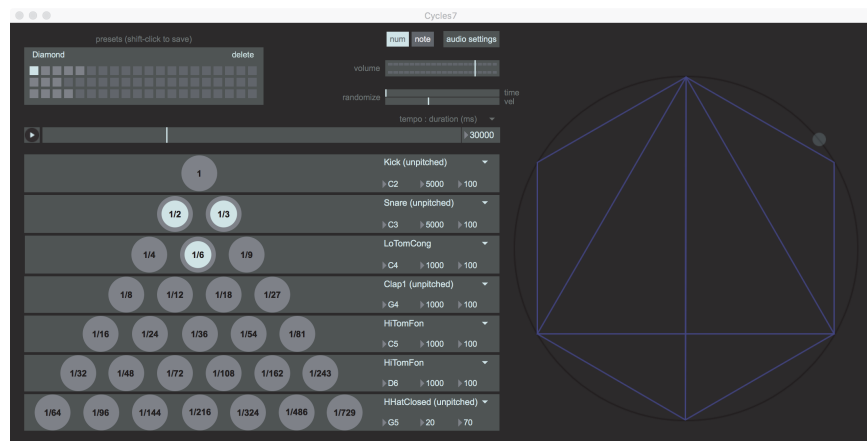


Fig. 9.8. Ski-hill graph (left) with the nodes mapped as fractions and a mod-6 cycle (right) illustrating a 3:2 hemiola; e.g., dotted semibreve and semibreve, or dotted minim and minim.

in XronoMorph, which are representations of a subset, can be interactively rotated. Getting students to engage and internalize these perception–action routines for transforming a subset may help them develop a more grounded understanding of the concept of equivalence. This form of interaction is supported in a computational domain and, in educational settings, may be best supported by touch-screen or free-form interfaces.

9.2.5.4. *Meter and ski-hill graphs*

Cohn [19] proposed an interesting representation of metrical structure designed to elucidate the multiple hierarchical levels that are inherent to meter in the common practice tradition, as illustrated in the triangular section of the software in Fig. 9.7.

The ski-hill graph is an isomorphic representation of meter. Each node represents a given pulse speed. The next node “South West” has a pulse that is doubled in speed; the next node “South East” has a pulse that is tripled in speed. This creates a hierarchy of possible relationships, where every metrical level (node on a row) can be subdivided into two beats and/or three beats. The pulse speed ratios between any two nodes is always invariant to translation: for example, two horizontal adjacent nodes are always in a 2:3 relationship; two nodes that are in a WSW relationship are always in a 3:4 relationship; and so forth.

This isomorphic structure (invariance with respect to translation) clearly represents the way that musical notation of meter is somewhat arbitrary (e.g., a given pulse level could be notated with a quarter note, an eighth note, or a half note, so long as all other levels make a corresponding change). It also emphasizes that metrical structures are “deeper” than is implied by standard time-signatures. For example, a time signature of $3/4$ tells us that each measure is divided into three quarter notes (crotchets), each of which is divided into two eighth notes (quavers). But it gives no information about longer metrical groupings (are measures grouped in twos or threes) nor about finer metrical subdivisions (are the eighth notes

divided into two or three)? This information is all explicitly — and isomorphically — visualized by the ski-hill graph.

The visualizations and sonifications provided by software such as XronoMorph and the SkiHill app offer numerous advantages over static versions of the same thing. One of these advantages is quite simple and intuitive: when reasoning about musical structures and ideas, one cares about how these more abstract relations translate into sonorous experience and how this makes one feel.

In a sense, learning music theory is about developing intuitive mappings between musical experiences and formal representations and operations. However, helping learners develop this connection between the formal and the experiential is challenging in the traditional classroom. In an ideal world, one would juxtapose the musical experience and the theoretical idea in close succession, and offer the learner opportunities to change an aspect of the model and see how this manifests in the musical experience. The ability to fluently make this transformation, however, is beyond the abilities of all but the most advanced students. One would either have to have very developed internal audiation abilities (to imagine in one's mind the resulting music accurately) or to have to perform this musical material on an instrument. These options are less immediate and, beyond a certain complexity of content, impossible.

Computer-based models allow one to simulate musical systems in a perceptually immediate way. This immediacy allows one to rapidly iterate between various settings of the model, facilitating hypothesis testing and creative explorations. Such ways of learning with computer-based models have become central to learning complex science concepts [38, 76, 81, 82] and similar gains, we argue, are to be found for mathematical music theory.

9.2.6. *Summary*

Despite their wide utility as models of musical structure, mathematical music theories can be challenging for non-mathematicians and, importantly in an educational context, children. However, when these

theories are instantiated in software applications with attractive visualizations and sonifications, with well-chosen parameterizations, the mathematics can be brought to life: made beautiful, relevant, and memorable. In the same way that computer-based models help children gain intuitions about challenging concepts such as calculus and complex systems [81], computer-based models of music may help children gain intuitions about concepts in mathematical music theory.

In summary, software visualizations and sonifications of mathematical music theory have at least three key benefits. First, they facilitate a deeper and broader understanding of music because they identify commonalities between different musical domains (e.g., pitch and time) and between different musical traditions (they are relevant to pop, rock, and non-Western musical traditions, as well as to common practice classical music); in this way, we can teach music with mathematics. Secondly, these mathematical principles act as high-level descriptors and manipulators of musical structure, which means they can serve as excellent tools to foster musical creativity and novelty. Thirdly, the musical realizations produced by the software can help to embody, visualize and sonify otherwise complicated mathematical ideas; in this way, we can also teach mathematics with music.

In the following section, music teacher Andrea Calilhanna gives us some invaluable “real world” insight into how software embodying two of the mathematical representations described above — the ski-hill graph and cyclical graphs of rhythm — have impacted on her teaching practice and on her students’ learning experiences.

9.3. From the Perspective of Andrea’s Teaching Studio

In 2015, I studied a semester of musical meter theory classes taught by Richard Cohn in Sydney, Australia. To begin, I wondered what I could possibly learn about meter after teaching for 30 years. Isn’t meter telling the student, for example, the piece is in 3/4,

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it's simple triple and to focus on the "strong, weak, weak" of the three crotchet beats per measure? This was what I had been taught, it was the information supplied by the secondary music classroom textbooks and theory books for exam preparation, and this is what would gain my students the marks they needed on their music examinations (my music studio work involves preparing students for piano and music theory examinations, tutoring for secondary high school Music 1 and 2, university auditions and preparation for tertiary music analysis and aural skills). By the end of the course, I had completely shifted in my understanding and teaching about meter and I experienced the realization that I had been teaching only a superficial understanding of meter. One of the main reasons for this leap in my thinking and practice was my introduction to visualizations of meter: the ski-hill and cyclical graph. In this section, I describe how I have used these visualizations to teach musical meter and rhythm, and how software animations and sonifications of these visualizations helped engage my learners.

Although Cohn [19] had originally intended the ski-hill graph as a way of graphically representing complex hemiolas in Beethoven's music, in 2015 I adapted it for teaching much younger students from primary and high school through to young adults. I find the ski-hill graph to be the most compact and efficient way for my students to graphically represent their hearing of the hierarchical relations between adjacent pulses that form meters. As discussed in the previous sections, this is because the ski-hill graph effectively uses spatial information to reflect the hierarchical organization of meter into pulses grouped in either 2s or 3s.

In a lesson, my approach is to ask students to map pulses they hear in the music to nodes on the ski-hill graph. Students map pulses in a relation of inclusion^b they hear forming duple relationships (duple meter) on the left pathways and pulses in a relation of

^b*Inclusion relations* refers here to sets of pulses which form meter through inclusion. The condition of inclusion requires that "each time point of the slower pulse is also a time point of the faster one". [20, class notes. Sydney].

inclusion they hear forming triple relationships (triple meter) on the right pathways. This is done first without traditional notation so as to build a foundation of metrical equivalence — the same metrical structure could be expressed in multiple redundant notations. From here, I prompt them to consider pulses at multiple levels to discuss how certain pulses are related to each other by either duple or triple inclusion relations. During this process, students are encouraged to articulate reasons why they choose to map pulses in the way they did and only look to the score for clues if necessary. Another appealing aspect of the ski-hill graph pedagogically is the ease with which the pulses can then be mapped onto other forms of metrical representation such as cyclical graphs. Cyclical graphs [21] then bring their own strengths in helping to demonstrate the cyclical and isochronous characteristics of meter — why repeat a pattern (meter) in a straight line if it can be represented once in a circle?

Aside from my frequent use of the ski-hill graph as a handwritten graphic representation of meter I introduced the SkiHill app (the prototype version discussed above) into my studio in 2017.^c The SkiHill app allows students to instantly produce sound and animations of the pulses they hear on nodes.

As an instrument for analyzing meter, the software version of the ski-hill graph allows a far more interactive and immediate experience through processes of listening, moving their bodies, and tapping/clapping to the beat. This is quite different to the more passive process of sitting with pen and paper, analyzing a score, or being told what to hear by a teacher. The student graphically represents their own hearing and temporal experience of meter from their own understanding.

^cThis occurred because, in 2015, I began to wonder if such an app existed but found nothing of its kind available. In 2016, I wrote to Richard Cohn and Andrew Milne and suggested the idea of developing a ski-hill graph app to help students learn more about meter. They both thought it was a good idea and thankfully for us we now have the world's first ski-hill graph app.

Analyzing meter through the SkiHill app means my students interact with the computer not as passive consumers of information [61, 64]; rather, they manipulate the computer so that their knowledge is transformed into symbolic representation. Thus, through this approach the computer becomes, in a sense, a “mediator” between information and knowledge [7]. To analyze meter in this way means students (and teachers) are then equipped to articulate their learning in many ways, such as through symbol, movement, language, and sound.

9.3.1. *Rohit conquers metric dissonance with ski-hill graphs*

One reason I started using ski-hill graphs was to help my students with musical performance through a deeper understanding of meter. For example, the Grade 3 piano examination piece “Abenddämmerung” (Evening Twilight) No. 3^d proved tricky for my student Rohit^e (13 years), who found he never performed measures 17–20 with stable timing and was baffled as to why this was the case. Together we studied the meter in “Abenddämmerung” by listening and drawing a ski-hill graph (see Fig. 9.9) and afterwards examined the score for clues (see Fig. 9.10, mm. 14–20).

Figure 9.9 illustrates the ski-hill graph we workshopped from Rohit’s hearing (reproduced here), which represents the pulses he heard forming two different meters in mm. 1–42. He then mapped it onto the ski-hill graph app to observe the sound, notation, fractions and mathematical proportions at work audibly and visually.

When mapping both meters together on the ski-hill graph (see Fig. 9.9) Rohit formed the hemiola he heard in mm. 17–20, which triggered indirect metric dissonance with the introduction of the

^d“Abenddämmerung” (Evening Twilight) No. 3 from *Notenbuch für Klein und Groß* Op. 138 by Stephen Heller measures 17–20.

^eReal names are not used.

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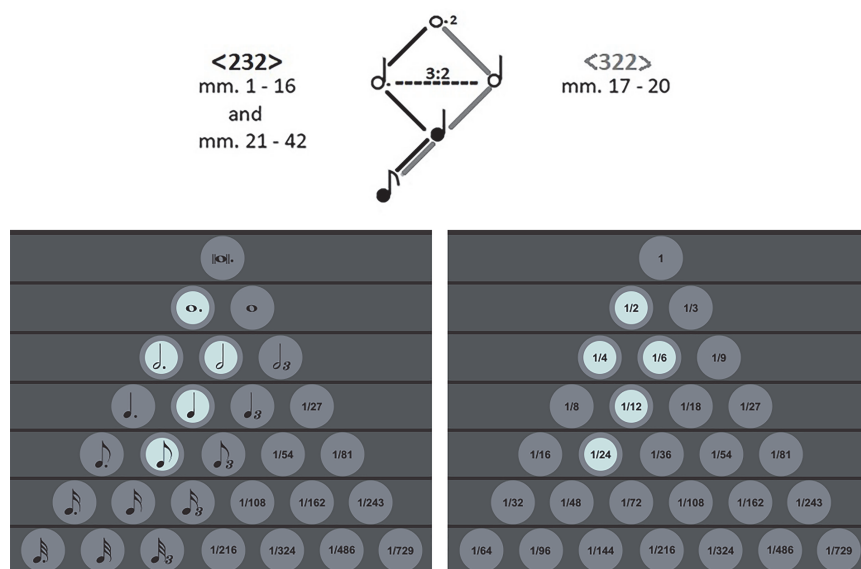


Fig. 9.9. The top figure illustrates a reproduction of my student's ski-hill graph graphically representing the meters $\langle 232 \rangle$ and $\langle 322 \rangle^f$ of "Abenddämmerung" (Evening Twilight) No. 3, which includes a hemiola. He then mapped it onto the SkiHill app to observe the sound, notation, fractions and mathematical proportions at work audibly and visually.

minim pulse (unlike direct metric dissonance, where the two pulses are heard simultaneously, here they are heard consecutively). He soon realized that this change in pulse was the cause of his timing problem. Through using the ski-hill graph, Rohit could easily hear and see how the two pulses (dotted minim and minim) form a ratio of 3:2 as shown by the horizontal dotted line (Fig. 9.9) and so were not in an inclusion relation (one pulse divides into the other by 2 or 3) hence the metric "conflict" or dissonance.

^fThese "deep meters" are parsimoniously represented with Cohn's *ordered set notation* [18], which show the inclusion relations of four metrical levels. For example, in $\langle 322 \rangle$ each pulse of the slowest metrical level contains 3 pulses of the next faster metrical level, which contains 2 pulses of the next faster level, which contains 2 pulses of the next faster level.

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Rohit and I analyzed how the composer had achieved this metric shift from $\langle 232 \rangle$ to $\langle 322 \rangle$ (both, confusingly, notated in 3/4 in conventional notation). The composer introduced metric displacement (syncopation) in measure 17 by subtracting a crotchet pulse (-1 ) and placing harmonic changes one unit early using restless V7 and I chords at onsets, which my student noticed led him to regroup the crotchets from sets of 3's to sets of 2's (notably, the crotchets were largely imagined rather than real to the hearing prior to measure 16). To subtract a crotchet pulse, the notated crotchet chord is placed a beat prior to the downbeat of measure 18 and tied over with the addition of the emphatic dynamic change *fp*, followed by two crotchet chords and a minim on the downbeat at measure 19; another set of tied crotchet chords (which to the hearing is effectively a minim pulse over mm. 19–20 with the dynamic *fp* followed by two crotchet chords).

Through these observations Rohit soon realized that his performance glitch would be solved if he shifted the set of pulses he entrained to from one set of pulses to another, as captured by his ski-hill graph. From our workshop, Rohit learned that there was more to meter in a piece than what was symbolically expressed in the notated time signature: the reality was far more dynamic and interesting. Figure 9.10 illustrates the evidence we gathered from listening, analyzing the notation and from pulses mapped on Rohit's ski-hill graph: the solid black arrows illustrate the dotted

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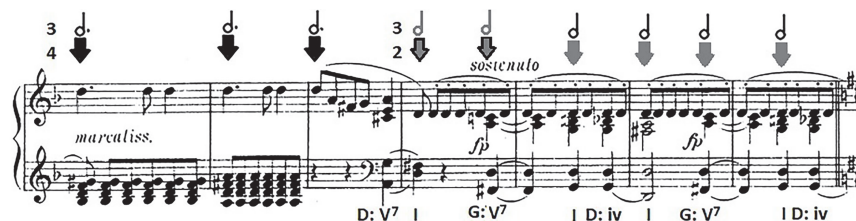


Fig. 9.10. “Abenddämmerung” (Evening Twilight) No. 3 from *Notenbuch für Klein und Groß* Op. 138 by Stephen Heller mm 14–20.

minim downbeat and grey arrows illustrate the minim pulse (the new “downbeat”). The grey arrows with black outlines are the pulses we labeled as metrically ambiguous (the pulses could be considered as either $3/4$ or $3/2$ at this point in the music). Since this realization, Rohit has never had a problem with the timing of these measures. It is always pleasing when students volunteer statements like, “Oh yeah! I get it!” and “I never thought about that before!”

Figure 9.11 shows Rohit listening to and mapping all of the pulses he hears in a new piece he is preparing for a Grade 4 piano exam “Bolero”.[§] Like before, he mapped the pulses he heard onto duple and triple pathways on the SkiHill app. From his previous lessons, he remembered that we often perceive pulses even when they don’t exist in the physical sound. For example, he “heard” crotchet and



Fig. 9.11. Rohit mapping his hand-written Ski-hill graph onto the SkiHill app. Photo credit: Andrea Calilhanna.

[§] “Bolero” No. 4 *11 Pezzi Infantile* Op. 35 by Alfredo Casella.

hypermetric pulses although there are no crotchet pulses and 2 or 4 measure hyperpulses notated. Through choosing adjacent pulses in inclusion relation on the SkiHill app, and in receiving instantaneous feedback of sound, Rohit gained a deeper understanding of the meter, including pulses which are not explicitly notated.

9.3.2. *Stella discovers mathematics helps her performances through understanding more about meter*

Like most children I've taught, Stella hadn't explicitly connected music with mathematics; yet, every time Stella works out the duple and/or triple relationships between the pulses that make up meter(s) she intuitively engages mathematical principles such as ratio, measurement, pattern, shape, multiplication, division, fractions, proportion, addition and maximal evenness. In my studio, we work through each of these principles as they arise relevant to the materials. I then encourage my students to familiarize themselves with the mathematics in the music, the music in the mathematics, and their embodiment of both music and mathematics, in order to learn more deeply about musical meter [8]. Stella is now quite at home identifying and demonstrating the hypermetric pulse in her piano pieces, which is a logical outcome of her understanding of meter as a hierarchy.

Stella's piece "The Banjo"^h is in pure duple meter and uses a repetitive pattern of quavers and minims to reflect the plucking of banjo strings. In measure 7, the pattern begins to change and, in measure 8, Stella repeatedly played the minims as crotchets. This was surprising considering no crotchets were notated on the score and no amount of counting and clapping seemed to help. I introduced Stella to the SkiHill app and asked her to map all of the pulses she could hear. She soon understood that the pulses she chose were related to

^hThe "Banjo" by Jennifer Trynes.

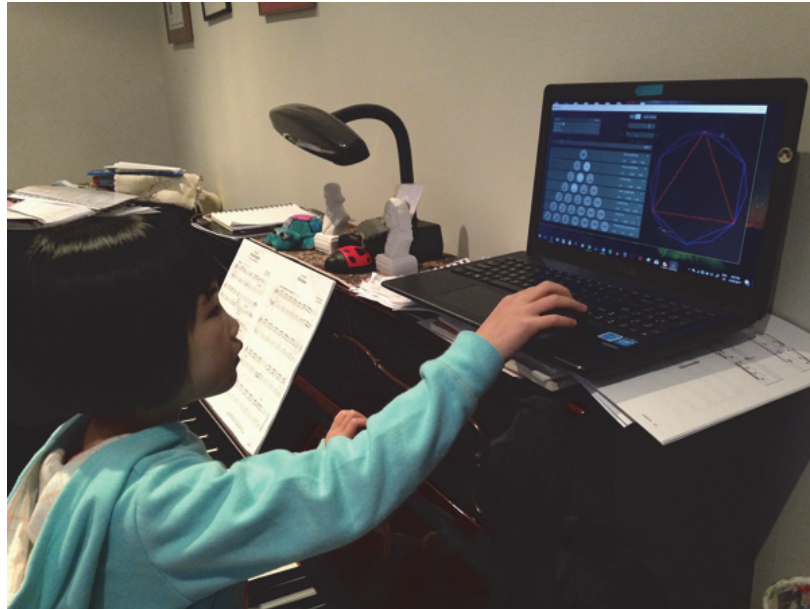


Fig. 9.12. Stella (aged 7) mapping all of the metrical pulses she hears in a piano piece she is preparing for an examination. Photo credit: Andrea Calilhanna.

each other in duple divisions and, when she related this to her difficult measure, she started to entrain to two or more pulses at a time, “filling in” the missing crotchet pulse and minim pulse. For Stella, this meant she stopped skipping a pulse causing the destabilizing effect on the performance of her preliminary exam piece. I asked Stella if this helped her in that one measure and she seemed pleased to say, “Yes it did!” (Figs. 9.12 and 9.13).

9.3.3. *Max the music turtle — more than meets the eye*

Max, see Fig. 9.14, is a toy turtle whose “job” it is to remind my very young students to curve their fingers and who occasionally spends a week with a student who isn’t well or who has worked hard at their lessons. Recently, I realized Max had another important message to



Fig. 9.13. The interactive ski-hill graph can also be used as a multi-pulse metronome, which helps students relate their theoretical understandings of meter to their playing and practice of music. Stella playing her piece as the SkiHill app sounds all of the pulses she is hearing. Photo credit: Andrea Calilhanna.

share with my students: the regular pattern of six sections on his shell could well serve as a mod-small cycle [21] to teach my students about cycles. I now use Max's shell to show my primary school students that they can place up to three pulses on Max's mod-6 cycle shell by placing colored discs to show pulses in inclusion relation making minimal and deep meters while listening to pulses in pieces in $3/4$, $3/8$ and $6/8$. My hope is Max will one day become an app for children where they can add pulses in inclusion relation and make sounds on a touch screen to make meters. Using mod-small cycles in ways such as this, and especially through using Milne's SkiHill app, makes for a very early introduction to the principles of set theory and, in particular, Richard Cohn's beat class set theory using mod-small cycles to help students understand meter more deeply.

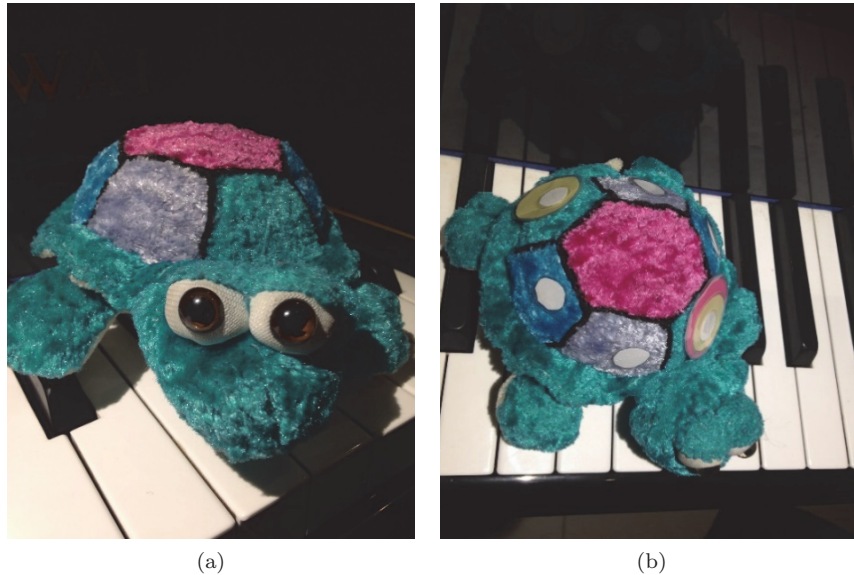


Fig. 9.14. (a) “Max” with a “mod-6 shell”; (b) “Max” modeling three pulses in one mod 6 cycle (equivalent to one measure of 6/8): the dotted minim (pink); dotted crotchet (yellow); quaver (white). Photo credit: Andrea Calilhanna.

9.3.4. Conclusion

In summary, through my own teaching practice, I have found both ski-hill and cyclical graphs an effective way for students to learn about meter and rhythm in an intuitive and non-intimidating way. Students were able to construct their metrical interpretations of the music, supported by software that scaffolded their growing understandings. My students have benefitted in so many ways, by gaining: a deeper understanding of the importance of musical meter and where meter is located; an ability to solve many problems involving analyzing and applying note values; awareness and use of division, proportion and counting when practicing new pieces; informed ability to identify pulses and articulate their choices of pulses; awareness of all the pulses that make up the meters they hear, including “imagined” pulses; understanding of hemiolas; comprehension of the cyclical nature of meter; metric equivalence; set theory; maximal evenness; awareness of notation as “quasi-arbitrary”; meter

as independent of the bar-lines and meter signatures; and increased confidence to perform, given their more secure metrical and rhythmic grounding.

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